

**Question:**  
Haversian system is seen in Answer Choices: A. Cortical bone B. Cancellous bone C. Teeth D. Nail

**Reasoning + Answer:**

<think> \*\*Finding reasoning paths: \*\* 1 . Bone tissue -> osteocytes ( cells ) -> haversian canal ( location of osteocytes ) 2 . Bone tissue -> osteon ( structure ) -> haversian canal ( component of osteon ) 3 . Bone tissue -> compact bone ( category ) -> haversian canal ( feature of compact bone ) \*\*Reasoning Process: \*\* 1 . \*\*Understanding the Haversian System:\*\* The Haversian system is a feature found in compact bone . It is composed of an osteon , which is a central hollow canal surrounded by a layer of osteocytes . 2 . \*\*Role of Osteocytes:\*\* Osteocytes are star-shaped cells located inside the bone tissue . They help maintain the structural and functional integrity of the bone by secreting substances such as calcium salts and osteocalcin . 3 . \*\*Relationship of Osteocytes to Osteons:\*\* Within the osteon , osteocytes are tightly packed in the center , surrounding the central canal , which is the haversian canal . This central canal is primarily a duct for blood vessels and nerves , but it also has some gap junctions . 4 . \*\*Classification of Compact Bone:\*\* Compact bone is a component of bone tissue that is continuous and dense , providing protection and structural support . It is differentiated from cancellous bone , which is less dense and honeycomb-like . 5 . \*\*Conclusion from Bone Tissue to Haversian System:\*\* Since the Haversian system is a structure within compact bone , it is logical to conclude that this system is found within the category of compact bone . \*\*Conclusion: \*\* The Haversian system , found within the bone tissue , is located in compact bone . </think> <answer> Cortical bone </answer>

